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The Hong Kong University of Science
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[Shuo Wang](#)

Education

Ph.D. in Resources and Environment
(Jun 2025)

Shandong University, China
GPA: 3.8/4.0

M.E. in Environmental Engineering
(Jun 2021)

Shandong University, China
GPA: 3.7/4.0

B.E. in Environmental Engineering
(Jun 2018)

Suzhou University of Science and
Technology, China
GPA: 3.5/4.0

Skills & Expertise

- Analytical chemistry
- Pure culture cultivation
- Field sampling and analysis
- Metagenomic data analysis
- Chromatography operation and maintenance
- Constructed wetlands and hydroponic system design

Curriculum Vitae

Shuo Wang

Postdoctoral researcher in environmental biotechnology at the Hong Kong University of Science and Technology (HKUST). Focused on using nanobubble technology to enhance phytoremediation for improving plant-based removal of environmental pollutants, including heavy metals and emerging contaminants.

Professional Experience

Aug 2025–Current

Postdoctoral Researcher

Department of Civil and Environmental Engineering
The Hong Kong University of Science and Technology,
Hong Kong
Research focus: Nanobubble-enhanced phytoremediation
of heavy metals and emerging contaminants
PI: Prof. Samir Kumar Khanal

Sep 2021–Jun 2025

Graduate Research Assistant

School of Environmental Science and Engineering
Shandong University, China
Research focus: Sulfur-driven phosphate bio-reduction and
its effect on enhanced phosphorus removal in constructed
wetlands
Supervisor: Prof. Zhen Hu

Sep 2018–Jun 2021

Graduate Research Assistant

School of Environmental Science and Engineering
Shandong University, China
Research focus: Characteristics of phosphine production in
constructed wetlands and its effect on phosphorus removal
of wetlands
Supervisor: Prof. Zhen Hu

Research Areas

- Phytoremediation
- Nanobubble technology
- Biological wastewater treatment
- Heavy metals and emerging contaminants removal

Publications

Journal Papers (Selected latest five years)

1. **Wang, S.**, Hu, H., Tanveer, M., Ji, M., Chai, W., Wu, H., Xie, H., & Hu, Z. (2024). Characteristics and mechanisms of phosphine production in sulfur-based constructed wetlands. *Water Research*, 256, 121639.
2. **Wang, S.**, Hu, Z., Zhang, J., Wu, H., Xie, H., Liang, S., Hu, H., & Jin, F. (2022). Formation of phosphine and its effect on phosphorus retention in constructed wetlands: Characteristic and mechanism. *Environmental Technology & Innovation*, 28, 102653.
3. **Wang, S.**, Li, X., Ji, M., Zhang, J., Tanveer, M., & Hu, Z. (2023). Is constructed wetlands carbon source or carbon sink? Case analysis based on life cycle carbon emission accounting. *Bioresource Technology*, 388, 129777.
4. Li, B., Hu, Z., Zhao, Q., Heng, J., **Wang, S.**, Khanal, S. K., Guo, Z., & Zhang, J. (2025). Enhanced fluoride removal in a novel magnesium-carbon micro-electrolysis constructed wetland through accelerated electron transport and anodic sacrifice. *Journal Of Hazardous Materials*, 492, 138062.
5. Lu, J., Yu, P., Zhang, J., Guo, Z., Li, Y., **Wang, S.**, & Hu, Z. (2024). Biotic/abiotic transformation mechanisms of phenanthrene in iron-rich constructed wetland under redox fluctuation. *Water Research*, 261, 122033.
6. Tanveer, M., **Wang, S.**, Ma, X., Yu, P., Xu, P., Zhuang, L., & Hu, Z. (2025). Enhancement of nitrogen transformation in media-based aquaponics systems using biochar and zerovalent iron. *Bioresource Technology*, 418, 131933.
7. Wu, W., Xu, F., Kong, Q., Hu, Z., **Wang, S.**, Zhang, J., & Ji, M. (2026). Enhancing the low-carbon performance for municipal wastewater treatment through high-rate activated sludge-three-stage constructed wetland system: Water quality, energy recovery and carbon emissions reduction. *Journal of Environmental Management*, 397, 128235.
8. Zhang, X., Heng, J., Zhao, Y., **Wang, S.**, Wang, Y., & Hu, Z. (2024). Effect of salinity on carbon sequestration in constructed wetlands and its functional mechanisms. *Bioresource Technology*, 391, 129915.

Book Chapters

1. Hu, Z., **Wang, S.**, Ji, M., Xu, X., & Tanveer, M. (2025). New experiences to enhance the treatment efficiency of constructed wetlands in cold climates. In *Emerging Developments in Constructed Wetlands* (pp. 579-594). Elsevier.

Awards and Honors

1. First Prize, Environmental Protection Science and Technology Award, China (2023)
2. Outstanding Graduate Cadre, Shandong University (2019)
3. Outstanding Graduate, Shandong Province (2021)
4. First Class Academic Scholarship, Shandong University (2022, 2023, 2024)
5. Outstanding Student, Shandong University (2019–2025)
6. Silver Prize, China International College Student Innovation Competition (2024)
7. Gold Prize, Shandong Province (China) College Student Innovation Competition (2024)
8. Champion and Runner-up, Basketball League, Shandong University (2018–2024)

Conference Presentations (* denotes presenter)

1. **Wang S.***, Study on the distribution and production mechanism of phosphine in wetlands, 4th National Conference on Water Treatment and Reuse of China, Jul 31-Aug 2, 2020, Xi'an, China. Oral presentation.
2. **Wang S.***, Hu Z., Sulfur-driven phosphate bio-reduction and its effect on enhanced phosphorus removal in constructed wetlands, 7th National Conference on Water Treatment and Reuse of China, Apr 14-16, 2023, Shanghai, China. Poster presentation.

Certifications

1. Lab Safety Training on Chemical, Biological and Fire Safety, Jan 2026
2. Responsible Conduct of Research (1-RCR), The Hong Kong University of Science and Technology, Nov 2025

Relevant Courses

Postgraduate Courses

(M.E. in Environmental Engineering)

- Engineering Ethics Attainment
- Water Treatment Engineering
- Air Pollution Control Engineering
- Engineering Mathematics
- Special Topic on Practice of Pollution Control Technology
- Frontiers Lectures of Environmental Engineering
- Analytical Technology and Method of Modern Apparatus
- Atmospheric Climate and Environment: Global Changes
- Advanced Wastewater Treatment

Postgraduate Courses

(Ph.D. in Resources and Environment)

- Modern Environmental Toxicology
- Membrane Water Treatment
- Major English
- Professional Practice
- Frontier Lectures

Undergraduate Courses

(B.E. in Environmental Engineering)

- Fundamentals of Environmental Science
- Fundamentals of Computer Science
- General Chemistry
- Advanced Mathematics
- Engineering Drawing Practice
- Physical Chemistry
- Production Internship
- Design of Water Treatment Structures
- Water Pollution Control
- Physical Pollution Control Engineering
- Air Pollution Course Design
- Solid Waste Treatment Course Design
- Environmental Impact Assessment
- Drainage Pipeline Engineering
- Pump and Pump Station Course Design
- Water Treatment Structure Course Design
- Physical Pollution Control Course Design